

Measurement of tidal volume

The air is inhaled through the nose. Here it is cleaned by hairs, preheated and moistened. Via the pharynx it goes on to the trachea with the larynx with the epiglottis. The trachea branches into the two bronchi, and these continue to branch, eventually ending in the alveoli. The alveoli are surrounded by capillaries (hair vessels). This is where gas exchange takes place: the blood absorbs oxygen and releases carbon dioxide.

With a "normal" breath, approx. 1/2l of air changes the lungs = breath volume. If you exhale to the maximum, another 1.7l can leave the lungs = expiratory reserve volume. About 1.3l remains as a residual volume in the lungs and cannot be measured with the spirometer. On the other hand, a few liters can be inhaled again by maximum inhalation = inspiratory reserve volume. The vital capacity is the total volume that can be maximally inhaled or exhaled.

The target value of the vital capacity for a person with L [cm] height and A age [years] is:

$$V_{K(Frau)} = [(51,5 - 0,08 * A) * L + 8 * A] - 4600$$

$$V_{K(Mann)} = [(61,5 - 0,33 * A) * L + 33 * A] - 5000$$

Task:

Measure each lung volume.

Material:

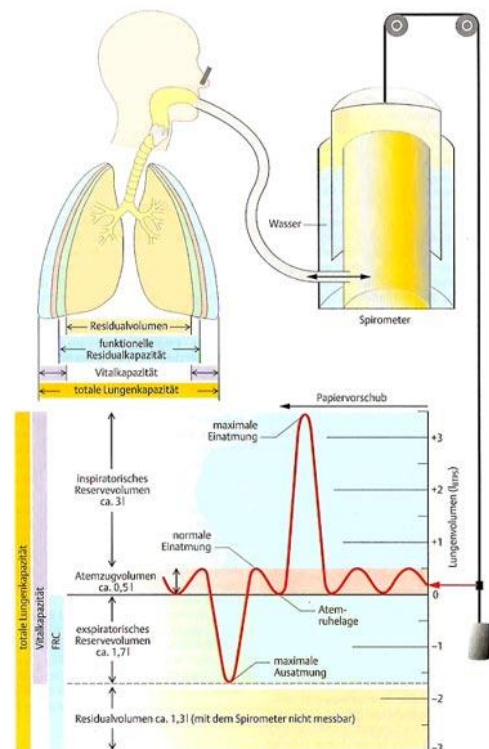
Handheld spirometer, alcohol

Execution:

- Clean the mouthpiece with alcohol
- Set the spirometer to 0 ("tare")
- Breathe in as deeply as possible and then blow as much air as possible into the spirometer

Evaluation:

Determining vital capacity. Comparison with a t-test of the target value with the actual value as well as with colleagues and with usual values.



Panoptic staining of blood smear

Acidic and basic dyes, mostly as color mixtures, are used to stain the blood smears. The possibility of differentiation consists in the different coloring behavior of the individual cell components towards different dyes. Erythrocytes and certain leukocyte granules are stained with different intensities by basic dyes.

We stain the blood cells according to Pappenheim, a panoptic stain, which is a combined May-Grünwald-Giemsa stain. The May Grünwald stain solution contains eosinic acid methylene blue in methanol. Methanol serves as a fixative. The Giemsa stain solution contains methylene azure, methylene violet, methylene blue and eosin dissolved in methanol and glycerol.

Task:

Two blood smears are to be prepared for each participant, one of which is stained

Material:

Microscope, (immersion oil), microscope slide, smear glass (cover slip), staining basin, tweezers

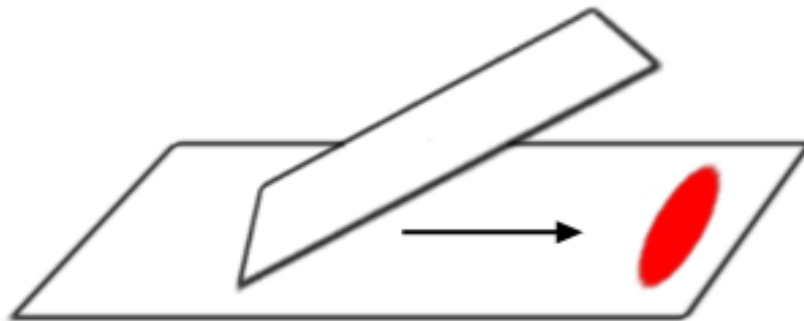
May Grünwald solution, Giemsa solution, water, fresh (!) blood

Execution:

CAVE: Wear gloves when working with blood!

CAVE: The staining solutions contain poisons such as methanol, so handle them with care. The staining solutions are also difficult to remove from hands and clothing. (Wear a work coat and gloves!)

To make a blood smear, a drop of blood is placed on the right side of a slide. The smear glass (which can be a second object slide, for example) is held between thumb and forefinger and placed at an angle of approx. 45° in the center of the object slide lying on the table.



Now you pull it towards the drop. When the drop has spread along the edge of the smear, the smear is quickly slid over the slide at an angle of 30° to the left so that an evenly thin film is drawn.

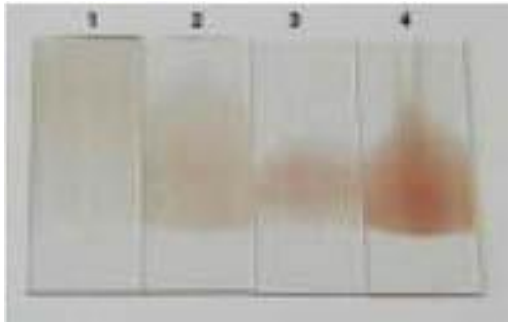


Then place the smear on the back of your hand and gently blow over it so that the film dries quickly and the cells are well preserved. Label the slide.

Caution: If you streak too quickly, the smear will be too uneven and too thin. If the smearing is done too slowly, there is a risk that the erythrocytes will take on a datura shape or stick together.

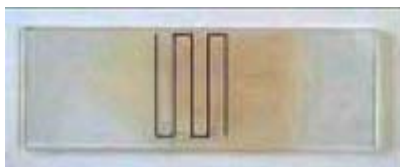
The best smears are stained using the Pappenheim method: To do this, place your preparations in a tub.

- May-Grünwald solution (fixation): 3min
- Transfer to a 1:1 mixture with distilled water: 1 min
- Pour off the color solution and immediately
- Place in a staining basin with freshly diluted Giemsa solution (10ml water + 10 drops Giemsa): 15min
- Rinse with distilled water and clean the underside of the glass
- Stand slides upright and air dry



- (1) too thin and too long
 (2) right!
 (3) Too short (too small drop?)
 (4) Too thick

Caution: Make sure that there is always enough staining solution on the slide to completely cover it. The colors must not dry out, it may be necessary to add color solution!



The unstained and stained blood smears are observed under the microscope and photographed. Count the leukocytes by guiding the field of view over the preparation in "snaking lines". Try to differentiate the different leukocytes (histological atlas or set of slides) and record them in a line chart.

	Mononukleäre Zellen		Granulozyten		
	Lymphozyten	Monozyten	Neutrophile	Eosinophile	Basophile
Liste					
Σ					
%					

Evaluation:

Enter the number of leukocytes (according to their type) or the percentage distribution. Include images of each type in your log. Compare your results with physiological values from the literature ("chisq.test"). Determine the size of the cells (measure several and average each group)

Determine the average size of the different types of leukocytes.

Conducting a blood group test

The ABO system was described by Karl Landsteiner in 1901, for which he received the Nobel Prize in Medicine in 1930. It is the most important blood group characteristic in blood transfusion and includes four different main groups: A, B, AB and O.

Blood group A has type A antigens present on the erythrocytes, blood group B has other type B antigens. People with blood group AB have both types of antigens, while blood group O has no antigens present. Conversely, people always have antibodies against the missing antigens, with blood group A they have antibodies against B and vice versa, with blood group AB no antibodies and with blood group O antibodies against both A and B. This is due to the intestinal bacterium (*Escherichia coli*), whose surface structure antigens on the erythrocytes is very similar. Between the 3rd and 6th months of life, the newborn develops antibodies against these surface structures of the bacteria if it is not itself a carrier of the similar antigens on the erythrocytes. In this case, since the immune system recognizes the surface structures of the bacteria as endogenous structures, it does not form any antibodies against them.

The points of attack of the antibodies are determined by the glycosylation of the blood proteins and lipids. A carrier of blood group A has antibodies that recognize and bind to the α -galactose (Gala for short) in the glycoside structure of the glycoproteins (blood group B). On contact, the erythrocytes agglutinate (clump together). However, blood group O lacks these antigens, so it does not cause agglutination and death in blood groups A and B. This makes carriers of blood group O with negative Rhesus factor universal donors, i. H. their blood can be used for carriers of all other blood groups.

The blood groups are determined by the genes A1/A2, B and O. The products of the O gene cannot be detected, ie the gene is silent. The products of the other genes are antigenically active glycoproteins. The two hereditary factors are located on the long arm of chromosome 9 (9q34). (wikipedia)

Task:

Determination of the blood group (ABO and D = Rhesus factor) with the help of antibodies

Material:

spot plate, toothpick

blood, (isotonic saline solution,) antibody sera A and B, Rh

Execution:

CAVE: Wear gloves when working with blood!

Pipette three (small) drops (approx. 20 μ l) of blood into three wells of the plate. Then a drop of Anti-A, Anti-B or Anti-Rh also in the wells. Mix them lightly with the toothpick. Agglutinations should be easily distinguishable from the unreacted areas.

Evaluation:

Determine their blood type, photograph the spot plate and discuss the result. Evaluate your group against the blood group distribution found in the Austrian population.

Determination of the erythrocyte count

In our case, the red blood cell count is carried out using a Neubauer counting chamber with a known area division and depth. For this purpose, the blood is diluted to 1:200 with an isotonic solution. The number of erythrocytes per liter of blood can be calculated from the mean number of erythrocytes in a counting square and the dilution.

Task:

In a blood sample, the number of erythrocytes (per liter), diameter (μm), surface area of an erythrocyte (μm^2) and all erythrocytes (m^2) in the body are to be determined and statistically evaluated.

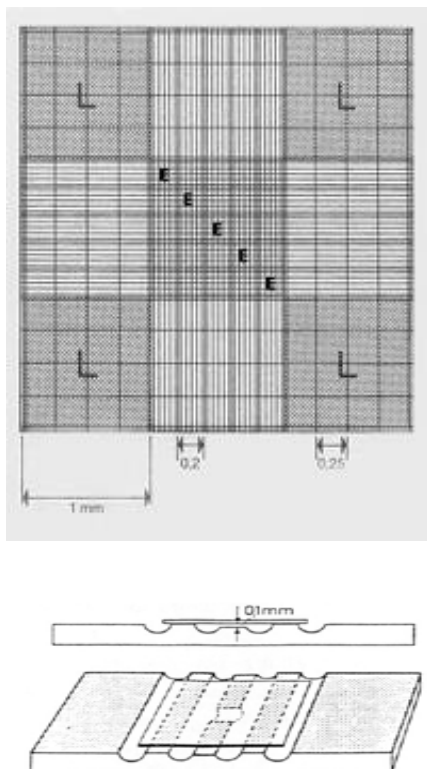
Material:

Microscope, counting chamber, coverslips, test tube, pipette

isotonic saline solution (0.92%), whole blood

Execution:

CAVE: Wear gloves when working with blood!



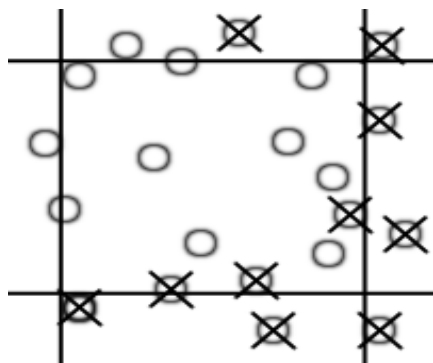
First of all, use the naked eye to get an idea of the position and structure of the counting fields in the Neubauer chamber. Clean the chamber and coverslip with ethanol and water - allow to air dry well.

The large corner squares (L) are used for white blood cell counts, the smaller ones labeled "E" are used for red blood cell counts.

The two side bars of the Neubauer chamber are moistened a little with a wet swab and the ground cover glass is pushed on under pressure (be careful not to break it!). If colored (so-called Newtonian) rings become visible under the cover glass and the cover glass does not come off when tilted, the two counting chambers created underneath have a defined volume.

Pipette 1990 μl of isotonic saline into an Eppendorf tube. Then measure a small drop of blood (10 μl) with a pipette and add it. The blood is thus diluted 1:200. Mix thoroughly but not too "violently" (slightly snap your finger on the tube).

Drops of the dilution are placed onto the edge of the coverslip and allow it to be drawn in by capillary forces. Too much liquid is sucked out with paper. Wait two minutes for the erythrocytes to settle. The cells are counted in at least five small squares (designated "E") under the microscope (with the aperture closed) at 100-400x magnification and the arithmetic mean is determined.



The count begins in the upper left corner of the counting field. Only those cells that are free within the square or touch the left or top side from the inside or outside or are on these sides are taken into account. Not counted are those erythrocytes that are either on the lower or right boundary line or touch it from the outside or inside.

The erythrocyte count is calculated as follows:

$$(\text{number of erythrocytes} \times \text{dilution}) / (\text{number of squares counted} \times \text{volume square}) [\text{cells/mm}^3]$$

Evaluation:

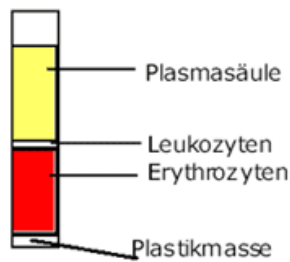
Calculate the number of erythrocytes per μl (or ml) of blood, compare this value with those found in the literature. Discuss the result.

Using the readings from all students, draw two histograms of the distribution of RBC counts in males and females. They should overlap to a significant extent. Calculate the two mean values and standard deviations and use the t-test ("t.test" in Excel) to determine differences in the mean values with an error probability of $\alpha=5\%$.

Determine the average size of an erythrocyte. Estimate the volume of an erythrocyte (see "Blood Numbers" document for data) and compare this value with the MCV value calculated by the "Hematocrit" group.

Reference values: males (4.8-5.9 million/ μl), females (4.3-5.2 million/ μl)

Determination of hematocrit



The hematocrit indicates the volume percentage of the blood cells in the blood. About 99% of this volume fraction are erythrocytes. The determination of hematocrit is a routine laboratory method. To accelerate sedimentation, heparinized (= anticoagulant, i.e. the blood remains liquid) capillaries filled with blood are centrifuged. The specific gravity of erythrocytes is higher than that of blood plasma. Therefore, the cells in unclotted blood sink. The hematocrit depends on the number and volume of the erythrocytes and is sex-specific.

Task:

The volume fraction of blood cells in the blood is to be determined using the microhematocrit method and expressed as a percentage.

Material:

Centrifuge, ruler (or normogram), micro hematocrit tube, capping kit, blood lancet, blood

Execution:

CAVE: Wear gloves when working with blood!

CAVE: The glass hematocrit tubes are very fragile!

Disinfect a fingertip with alcohol. Prick a fingertip with the blood lancet and use the capillary effect to fill the tube as much as possible (preferably 4/5) (keep the tube as horizontal as possible). The lower end of the capillary is sealed with the kit (pierce the tube into the kit mass, being careful not to break the glass). Then place in opposite (15 ml) tubes of the Hettich centrifuge (you normally use your own hematocrit centrifuge).

Centrifugation: 3 min at 2000 rpm (ask the course assistant)

Then the capillary tubes are evaluated using the normogram (if this is not available, the liquid columns can also be measured with a ruler): The capillary tube is pushed over the normogram until the entire liquid column is just between the upper and lower limit line (100 % and 0%).

Evaluation:

Record and discuss the results. Compare your data with those of other internship participants and create a small statistic (histogram), separated by males and females. Calculate the two mean values and standard deviations and use the t-test ("ttest" in Excel) to determine differences in the mean values with an error probability of $\alpha=5\%$.

Identify the secondary red cell counts:

Mean corpuscular Hb concentration MCHC = hemoglobin content/hematocrit value

Mean corpuscular hemoglobin MCH = hemoglobin content/erythrocyte count

Mean cell volume MCV = hematocrit value/erythrocyte count

Reference values: male (40-54%), female (37-47%)

Determination of methemoglobin in the blood

Hemoglobin is the blood's oxygen-carrying protein and has a quaternary structure of 2 alpha and 2 beta chains. Each of these peptides contains a porphyrin ring with a central iron atom. This can be demonstrated chemically.

The hemoglobin content of the blood is determined using the hemoglobin cyanide method. Hemoglobin (Hb) is oxidized to hemoglobin by potassium hexacyanoferrate III (KHCIII). This reacts with potassium cyanide to form hemoglobin cyanide, which has a high level of stability and is well suited for photometric concentration determination. In addition to hemoglobin, oxyhemoglobin, CO-hemoglobin and hemoglobin are also converted into hemoglobin cyanide by the so-called transformation solution and thus recorded quantitatively.

Task:

The hemoglobin content of the blood (g Hb per dl blood) is to be determined using the hemoglobin cyanide method and the reflotron, and the two methods are to be compared with one another.

Material:

Photometer, cuvettes, test tubes, pipettes, parafilm, reflotron+test strips

Transformation solution (already prepared; KHC(III) 0.2 g/l, KCN 0.2 g/l), blood

Execution:

CAVE: Wear gloves when working with blood!

CAVE: Potassium cyanide is a serious respiratory poison!

1) 5 ml of the transformation solution and 20 µl of blood are placed in a test tube. Seal the tube with Parafilm and mix by tilting the test tube. Fill the solution into a cuvette and wait at least 3 minutes (in the dark). For reference (blank) use pure transformation solution. Measure in the photometer at 546nm.

2) Put 30 µl of whole blood on an Hb test strip and measure in the reflotron.

Evaluation:

The hemoglobin concentration is derived from the measured extinction (E_λ) according to the Lambert-Beer law: $c = E_\lambda / \epsilon_\lambda \cdot d$

ϵ_λ = molar extinction coefficient

c: concentration [mol/l]

d = cuvette pathlength 1cm

The Hb concentration of the sample is in [g/dl], taking into account the molecular weight for hemoglobin (MGhb: 64,458), the molar extinction coefficient ($\epsilon_{546} = 44,000$) and the ratio of the final volume of the preparation ($E_v = 5.02\text{ml}$).) and the sample used ($P_v = 20 \mu\text{l}$)

$$c = E_{546} \cdot \text{MGhb} / \epsilon_{546} \cdot E_v / P_v \cdot 1/d \text{ [g/l]} = E_{546} \cdot 36.8 \text{ [g/dl]}$$

Compare the manually measured value with that of the Reflotron and compare the males's data with that of the females. Calculate the two mean values and standard deviations and use the t-test ("t.test" in Excel) to determine differences in the mean values with an error probability of $\alpha=5\%$.

Measurement of osmotic erythrocyte resistance

Given sufficient time, virtually any molecule will diffuse across a plasma membrane in the direction of the concentration gradient. However, the speed at which this happens varies enormously, depending on the size of the molecule, but mainly on its hydrophobicity. For example, plasma membranes are 10⁹ times more permeable to water than even small ions such as K⁺ or Na⁺. Since the plasma membrane is weakly permeable to water, this slowly flows into or out of the cell depending on its concentration gradient (=osmosis). If the concentration of dissolved particles in the external environment is lowered (hypotonic solution), this causes an overall movement of water into the cell (it swells). Conversely, a hypertonic solution causes the cell to leak water (it shrinks). How long an erythrocyte can withstand an influx of water before it finally bursts (osmotic hemolysis) is an expression of its physiological functionality and is therefore of great diagnostic interest, because there are diseases in which the osmotic resistance of the erythrocytes is increased (e.g. thalassemia, sickle cell anemia, liver damage, etc) and those in which it is reduced (e.g. spherical cell anemia, hemolytic anemia).

Task:

Determining erythrocyte osmotic resistance

Material:

15 ml centrifuge tube, pipette, 1.0% NaCl solution, distilled water, blood

Execution:

CAVE: Wear gloves when working with blood!

Prepare a NaCl concentration series from 0.2-0.9% in 0.1% increments in individual Falcon tubes (labeled!). To do this, add the following volumes from the pipetting scheme:

Endkonz	1% NaCl [ml]	Aqua dest [ml]	Endvolumen
0,80	2,4	0,6	3 ml
0,70	2,1	0,9	3 ml
0,60	1,8	1,2	3 ml
0,50	1,5	1,5	3 ml
0,45	1,35	1,65	3 ml
0,40	1,2	1,8	3 ml
0,35	1,05	1,95	3 ml
0,30	0,9	2,1	3 ml
0,20	0,6	2,4	3 ml

Mix the blood by tilting the blood tube several times and use the pipette to add 100 µl of blood to each centrifuge tube. After sealing, shake it vigorously and leave it for about 10 minutes. Then centrifugation for 5 minutes at 1000 rpm (the course supervisor will help).

Measure the supernatant using the hemocyanide method!!!

To do this, mix 900 µl of transformation solution and 100 µl of supernatant.

Evaluation:

Look for those tubes in which the saline solution is no longer completely colorless but is already slightly reddish. This is the concentration at which the first erythrocytes burst. Also add the tube that has no red cells at all at the bottom of the tube. Record and discuss your findings.

Reference range for human erythrocytes: 0.30 - 0.46.

Protein Concentration Determination

The dye Coomassie blue reacts with basic amino acids. The absorption maximum shifts from 470 nm to 595 nm. By measuring the absorbance at this wavelength, the concentration of an unknown protein solution can easily be determined, provided the protein used for the calibration and the unknown protein have approximately the same basic amino acid content. Bovine serum albumin (BSA) has established itself as the internationally recognized standard for calibration.

Task:

Determine the blood protein concentration using the Bradford Assay.

Material:

Photometer, cuvettes, pipettes, 1.5ml reaction vessels (= "Eppi")

Coomassie reagent (20mg Serva Blue G in 25ml 96% ethanol and 50ml p-phosphoric acid (85%), total 500ml), 4mg/ml BSA, water, whole blood or serum

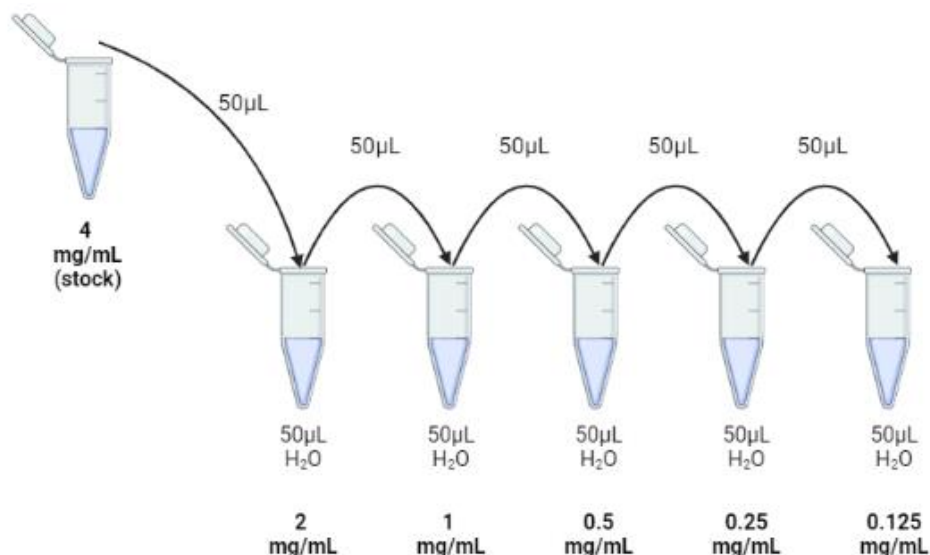
Execution:

CAVE: Wear gloves when working with blood!

In order to determine the amount of protein based on the measured absorption values, a calibration line is created with BSA (bovine serum albumin):

To do this, prepare a dilution series (50 μ l for each (4/2 /1 /0.5 /0.25 /0.125 mg/ml) from a 4 mg/ml BSA solution in deionized water. To do this, proceed as follows:

- Pipette 50 μ l of deionized water into an Eppi
- Add 50 μ l of the 4mg/ml solution to the first Eppi, mix well -> 2mg/ml
- Withdraw (with a new pipette tip) 50 μ l and put it into the next Eppi. Mix well -> 1mg/ml
- Repeat for the remaining vessels



Dilute the blood serum or whole blood 1:50 and 1:200 with water, ie

- 100µl serum + 4.9ml deionized water. respectively
- 50µl serum + 9.95ml deionized water

Fill 8 cuvettes with 1ml Coomassie staining reagent each. Then add 20µl each of the BSA protein dilution or the diluted samples and mix thoroughly. Allow the batch to incubate ≥ 10 min at room temperature. The absorption is measured at 595 nm in a spectrophotometer (1 ml Coomassie staining reagent is used as a reference).

Evaluation:

Draw a diagram in Excel with the absorbance (y-axis) versus the BSA concentration (x-axis) = calibration line and use it to determine the protein concentration of the serum. Describe the values of all subjects together and compare them statistically with the reference value of 7g/l (e.g. in Excel with the function `g.test()`).

Blood cell settling rate

The specific gravity of erythrocytes is higher than that of blood plasma. Therefore, the cells in unclotted blood sink. The speed at which this happens depends on the red blood cells' ability to cluster. This is promoted by certain plasma proteins, which weaken the suspension-stabilizing electric fields between the negatively charged erythrocytes. Accordingly, an increased rate of sedimentation (ESR) is observed in infections, inflammations and tissue breakdown.

BCS determination typically requires 1.6 mL of blood mixed with 0.4 mL of sodium citrate (which inhibits clotting). This mixture is filled into a 20 cm long vertical tube (internal diameter 2.5 mm). After an hour, it is read how much the blood cells have dropped. A second reading of the reduction after another hour is usual, but this two-hour value does not provide any additional information.

Reference / normal values after one hour is <15 mm for males and <20 mm for females.

Task:

Determining the blood cell settling rate

Material:

Westergren sedimentation pipettes or hematocrit tubes, sedimentation stand, blood, 10% EDTA solution or 3.2% trisodium citrate solution.

Execution:

CAVE: Wear gloves when working with blood!

For capillary blood, fill a hematocrit tube 4/5 full with EDTA blood. Mix whole blood 1:5 with EDTA solution. Measure the total height in the tube and the plasma column after 30, 60, 90, 120 and 150 min.

Evaluation:

Calculate the reference values compared to the values of hematocrit tubes given above. Create a graph of plasma column height versus time. Compare the measurements of the participants among themselves, between males and females (t.test) and with the reference values (g.test).

Comparison blood pressure monitors

The auscultatory (auscultare = lat. to listen) determination of the systolic and diastolic blood pressure is possible by using a manometer cuff. By applying and inflating the pressure cuff, the brachial artery is compressed from the outside in such a way that the systolic pressure wave coming from the heart is interrupted. While the external pressure is slowly being released, the pulse in the vessel beyond the point of compression can be heard again via a stethoscope at the moment when the blood pressure during systole is higher than the pressure of the cuff (systolic blood pressure). The blood can only pass through the clamped point of the brachial artery during a short period of systole, after which the arterial walls are compressed again (Korotkov pulse sounds are audible).

As external pressure continues to decrease, the pulse sounds can be heard as long as the artery remains occluded to flow during diastole. If the cuff pressure falls below the diastolic blood pressure, the pulse noise disappears completely. The manometer reading displayed corresponds to the diastolic blood pressure.

Task:

Determination of systolic (during ventricular contraction) and diastolic (during ventricular relaxation) blood pressure bloodless according to Riva-Rocci at rest and after physical exertion. Comparison of the manual and machine method.

Material:

Stethoscope, manual and automatic blood pressure monitor

Execution:

The rubber cuff of the blood pressure device is placed tightly (but not too tightly) around the released left upper arm (the elbow must remain free). The cuff pressure is to be brought to approx. 170 mmHg (25 kPa) with the valve screw closed (the brachial artery is compressed from the outside in such a way that the systolic pressure wave coming from the heart is interrupted), at the same time the pulse is felt on the wrist or in the Elbow checked with stethoscope.

The cuff pressure is then reduced by slowly opening the valve screw until you can feel the pounding pulse on your wrist or hear it in the stethoscope (Korotkov sound; corresponds to the maximum systolic blood pressure). If the pressure is then further slowly reduced, the throbbing pulse becomes less noticeable or audible in the stethoscope = minimal diastolic blood pressure. After the determination, the pressure is released completely using the valve screw and the cuff is removed.

Repeat this measurement after about 3 minutes alternately with the manual or automatic blood pressure monitor.

After a few minutes, do 30 squats and repeat the measurements!

Evaluation:

Record the determined data in tabular form. Calculate means and standard deviations. Using a t-test, compare the two methods or at rest and after exercise. Discuss the values!